

Reinforcement Learning Basics

Shusen Wang

Stevens Institute of Technology

<http://wangshusen.github.io/>

A little bit probability theory...

Random Variable

- **Random variable**: unknown; its values depend on outcomes of random events.

*Random
Variable*

*Possible
Values*

*Random
Events*

Probabilities

$$X = \begin{cases} 0 \\ 1 \end{cases}$$



$$\mathbb{P}(X = 0) = 0.5$$

$$\mathbb{P}(X = 1) = 0.5$$

Random Variable

- **Random variable**: unknown; its values depend on outcomes of random events.
- Uppercase letter X for a random variable.
- Lowercase letter x for an observed value.
- For example, I flipped a coin 4 times and observed:
 - $x_1 = 1$,
 - $x_2 = 1$,
 - $x_3 = 0$,
 - $x_4 = 1$.

Probability Density Function (PDF)

- PDF provides a relative likelihood that the value of the random variable would equal that sample.

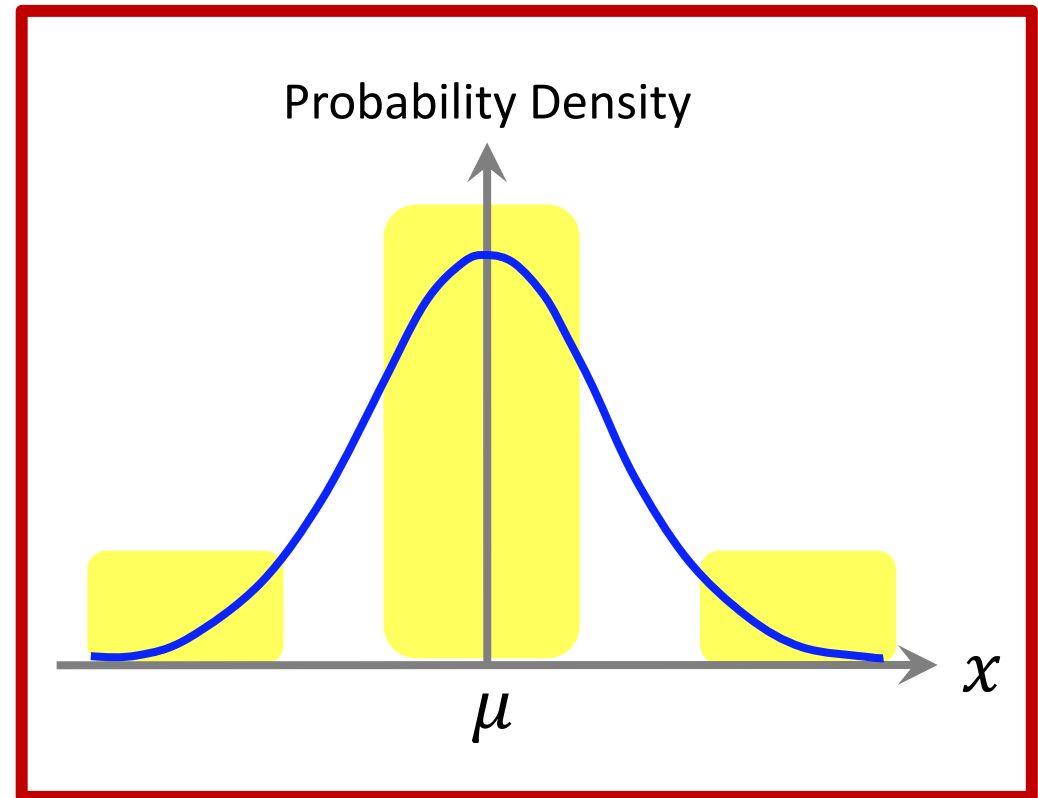
Probability Density Function (PDF)

- PDF provides a relative likelihood that the value of the random variable would equal that sample.

Example: Gaussian distribution

- It is a continuous distribution.
- PDF:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$



Probability Mass Function (PMF)

- PMF is a function that gives the probability that a discrete random variable is exactly equal to some value.

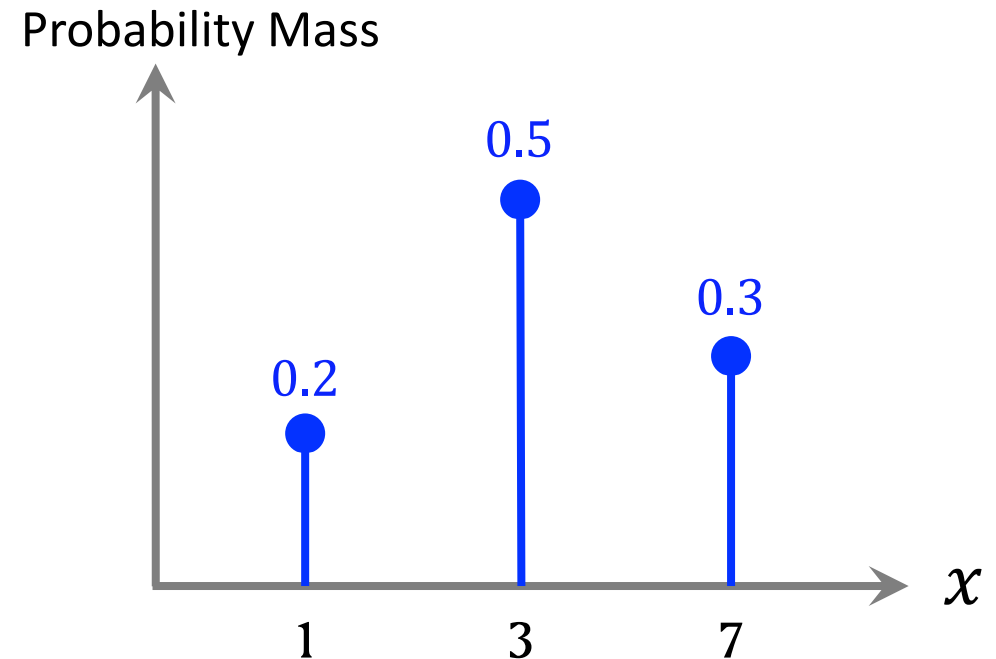
Probability Mass Function (PMF)

- PMF is a function that gives the probability that a discrete random variable is exactly equal to some value.

Example

- Discrete random variable: $X \in \{1, 3, 7\}$.
- PDF:

$$\begin{aligned}p(1) &= 0.2, \\p(3) &= 0.5, \\p(7) &= 0.3.\end{aligned}$$



Probability Mass Function (PMF)

- PMF is a function that gives the probability that a discrete random variable is exactly equal to some value.

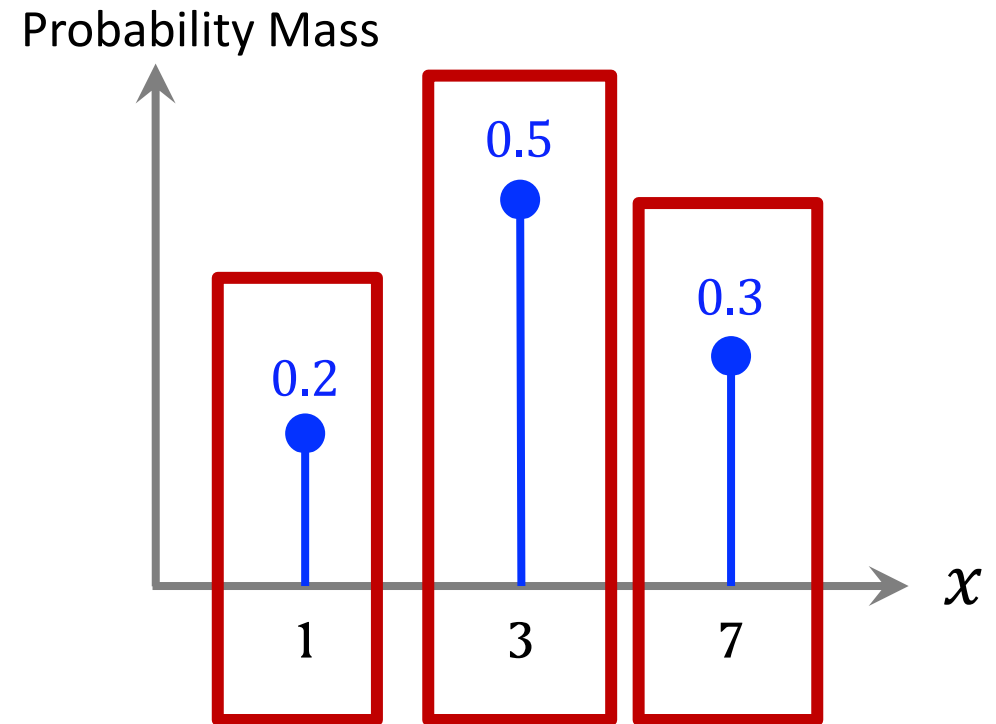
Example

- Discrete random variable: $X \in \{1, 3, 7\}$.
- PDF:

$$p(1) = 0.2,$$

$$p(3) = 0.5,$$

$$p(7) = 0.3.$$



Properties of PDF/PMF

- Random variable X is in the domain $\mathcal{X}$.

- For continuous distributions,

$$\int_{\mathcal{X}} p(x) dx = 1.$$

- For discrete distributions,

$$\sum_{x \in \mathcal{X}} p(x) = 1.$$

Expectation

- Random variable X is in the domain $\mathcal{X}$.
- For continuous distributions, the expectation of $f(X)$ is:

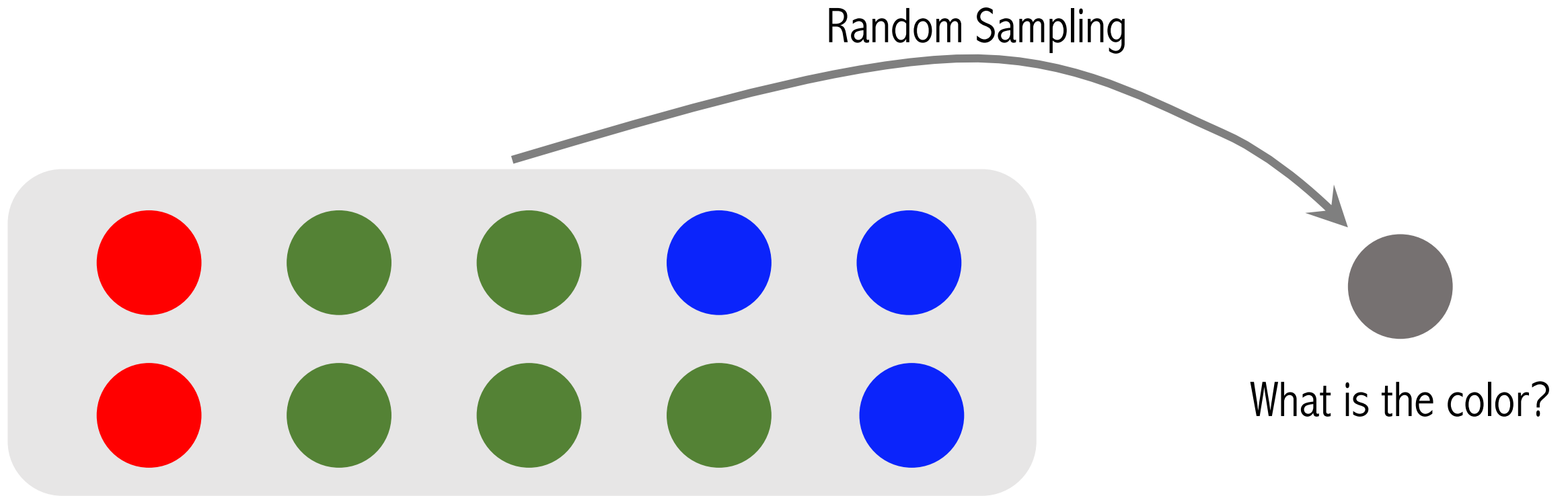
$$\underline{\mathbb{E} [f(X)]} = \underline{\int_{\mathcal{X}} p(x) \cdot f(x) dx}.$$

- For discrete distributions, the expectation of $f(X)$ is:

$$\underline{\mathbb{E} [f(X)]} = \underline{\sum_{x \in \mathcal{X}} p(x) \cdot f(x)}.$$

Random Sampling

- There are 10 balls in the bin: 2 are red, 5 are green, and 3 are blue.



Random Sampling

- Sample red ball w.p. 0.2, green ball w.p. 0.5, and blue ball w.p. 0.3.

Random Sampling

- Sample red ball w.p. 0.2, green ball w.p. 0.5, and blue ball w.p. 0.3.

```
from numpy.random import choice
```

```
samples = choice(['R', 'G', 'B'], size=100, p=[0.2, 0.5, 0.3])
```

```
print(samples)
```

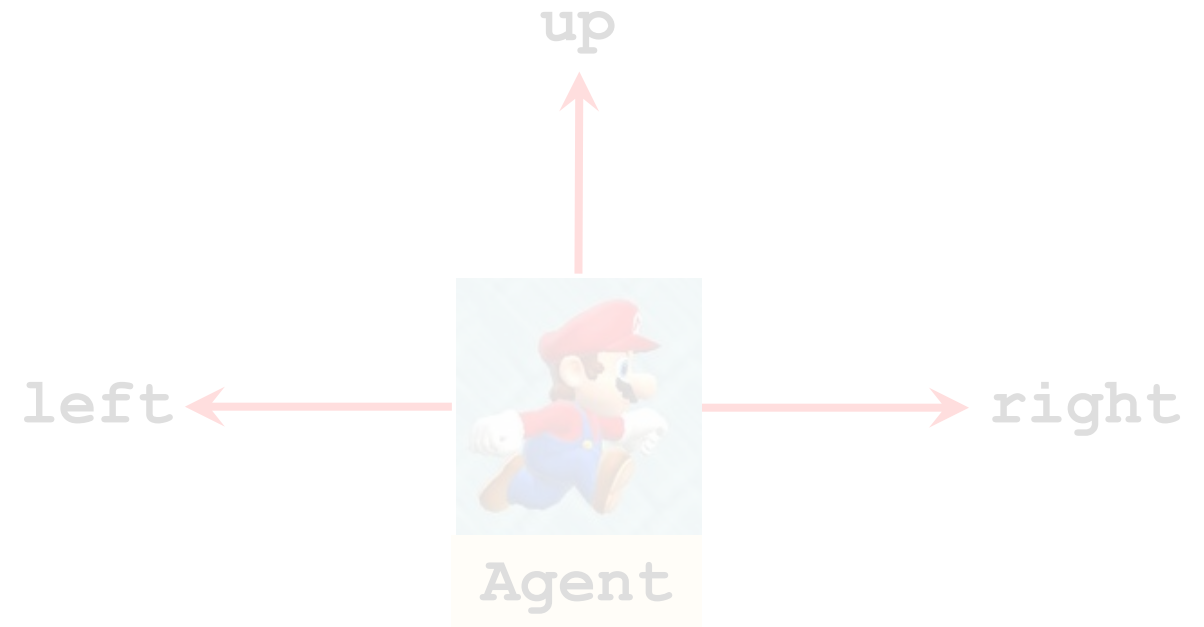
```
[ 'R'  'G'  'R'  'R'  'R'  'R'  'B'  'B'  'B'  'G'  'G'  'B'  'G'  'B'  'B'  'G'  'B'  'G'
  'B'  'B'  'G'  'B'  'G'  'B'  'B'  'G'  'B'  'B'  'G'  'B'  'G'  'G'  'G'  'G'  'G'  'B'
  'B'  'B'  'B'  'B'  'B'  'G'  'G'  'B'  'R'  'R'  'B'  'R'  'B'  'G'  'R'  'G'  'R'  'G'
  'R'  'R'  'B'  'G'  'G'  'G'  'B'  'R'  'G'  'B'  'G'  'R'  'G'  'G'  'G'  'B'  'B'  'R'
  'G'  'G'  'B'  'B'  'R'  'B'  'B'  'B'  'R'  'B'  'G'  'B'  'R'  'B'  'R'  'G'  'B'  'R'
  'B'  'B'  'G'  'G'  'G'  'R'  'R'  'B'  'R'  'G' ]
```

Terminologies

Terminology: state and action

state s (this frame)

Action $a \in \{\text{left}, \text{right}, \text{up}\}$

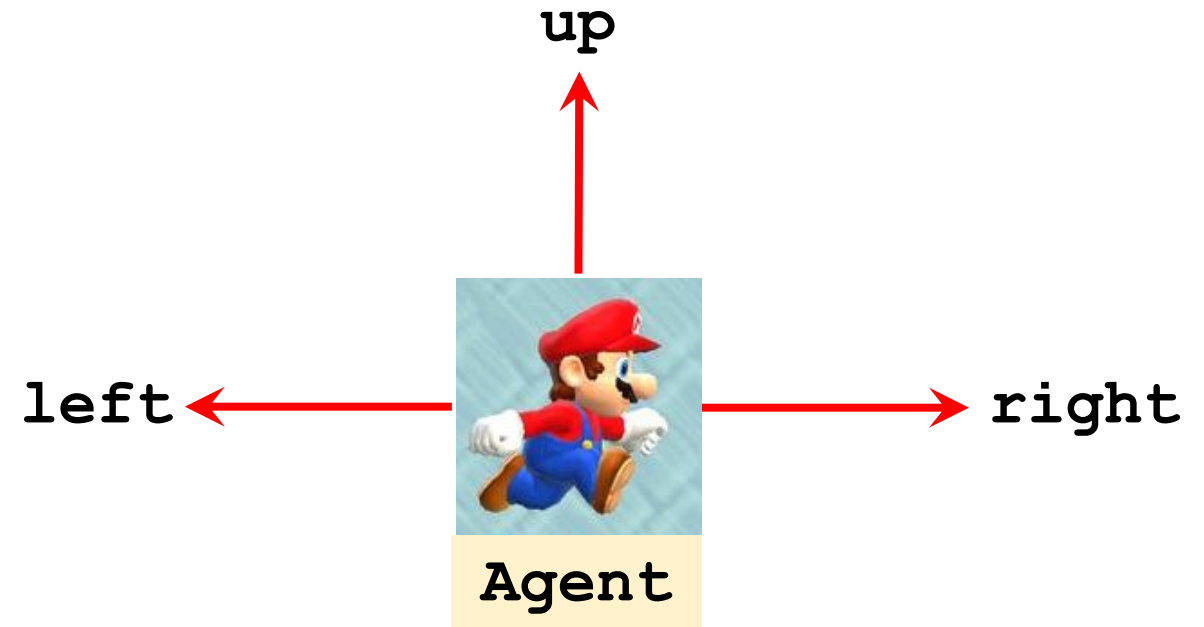


Terminology: state and action

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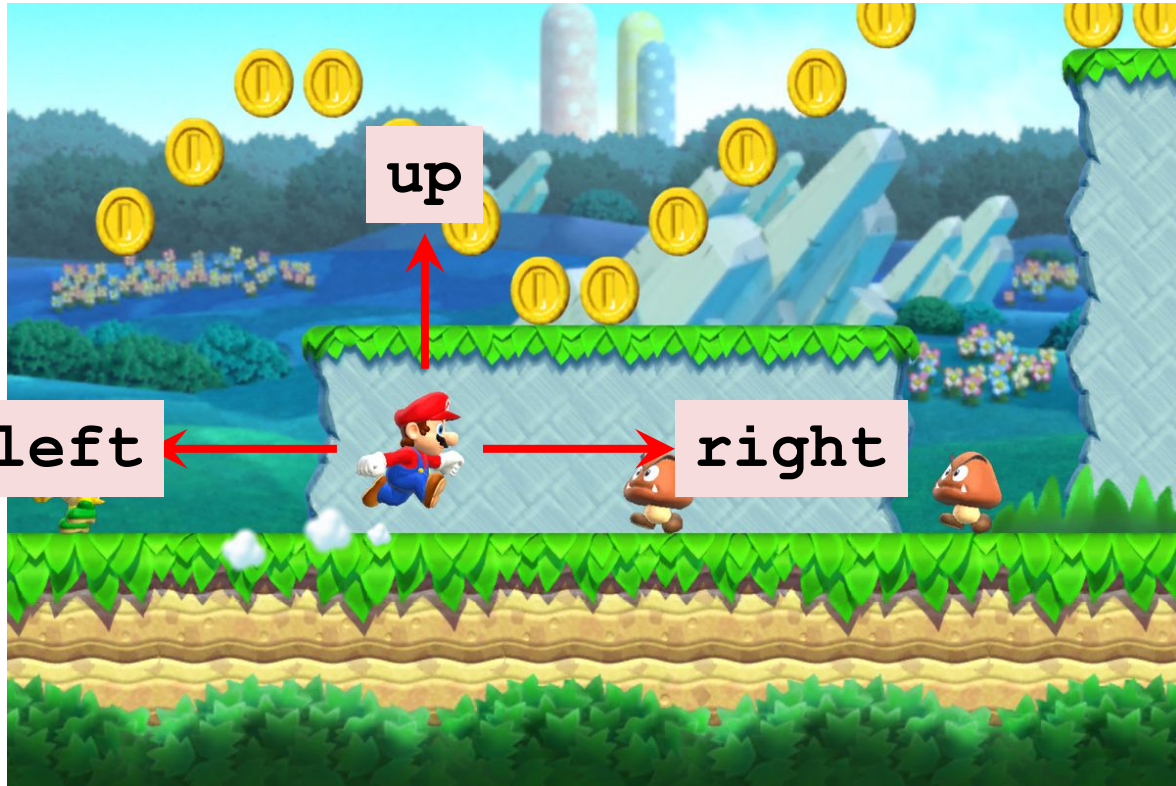


Terminology: policy

policy π

- Policy function $\pi: (s, a) \mapsto [0,1]$:

$$\pi(a | s) = \mathbb{P}(A = a | S = s).$$

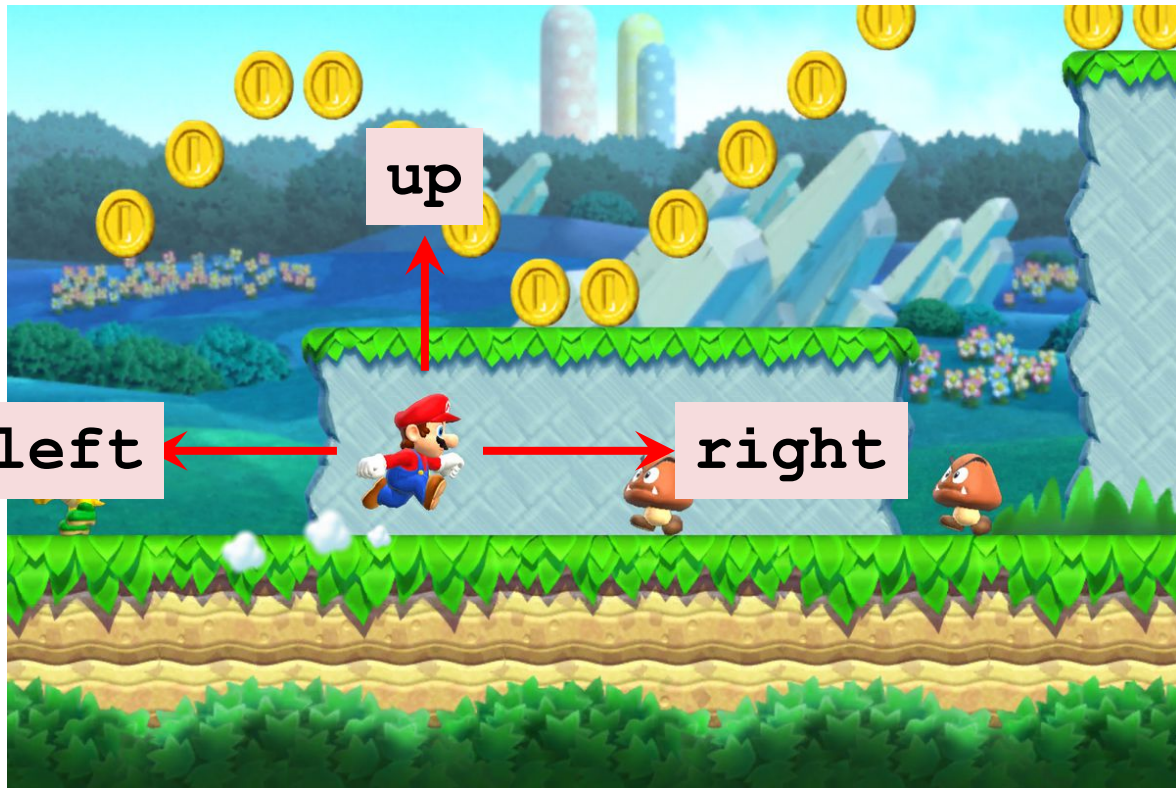


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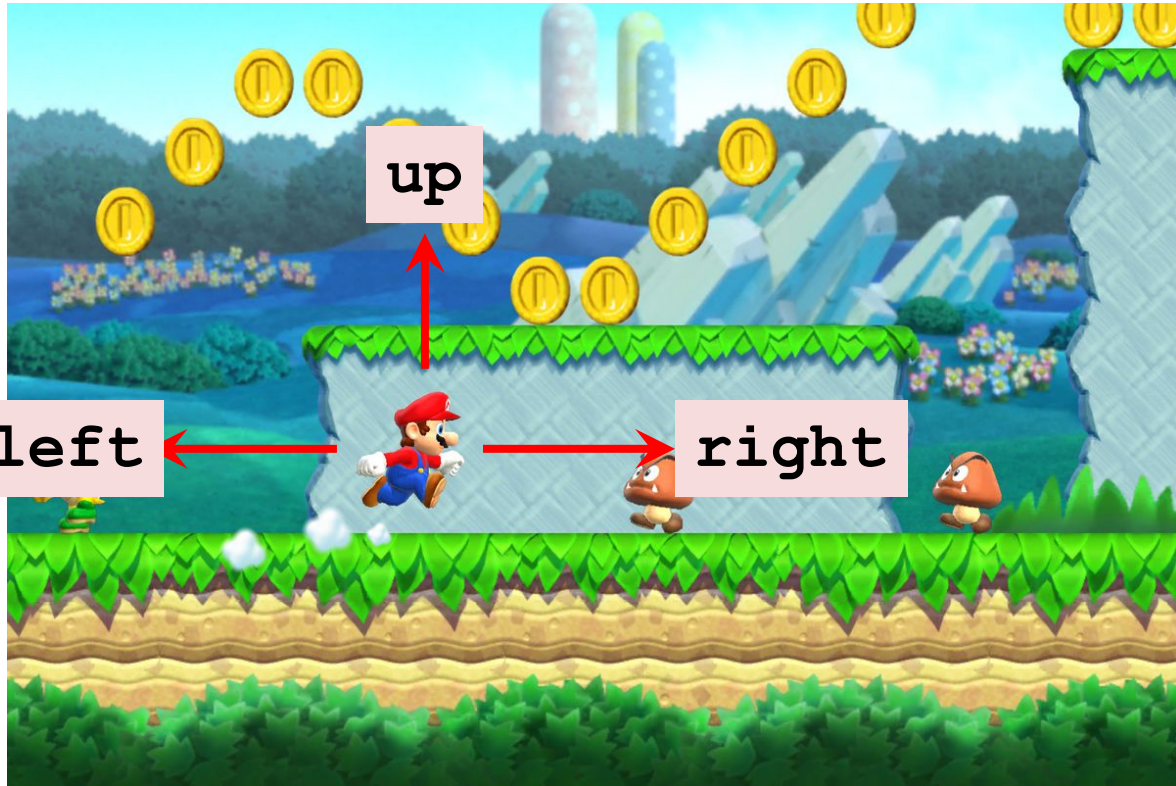


Terminology: policy

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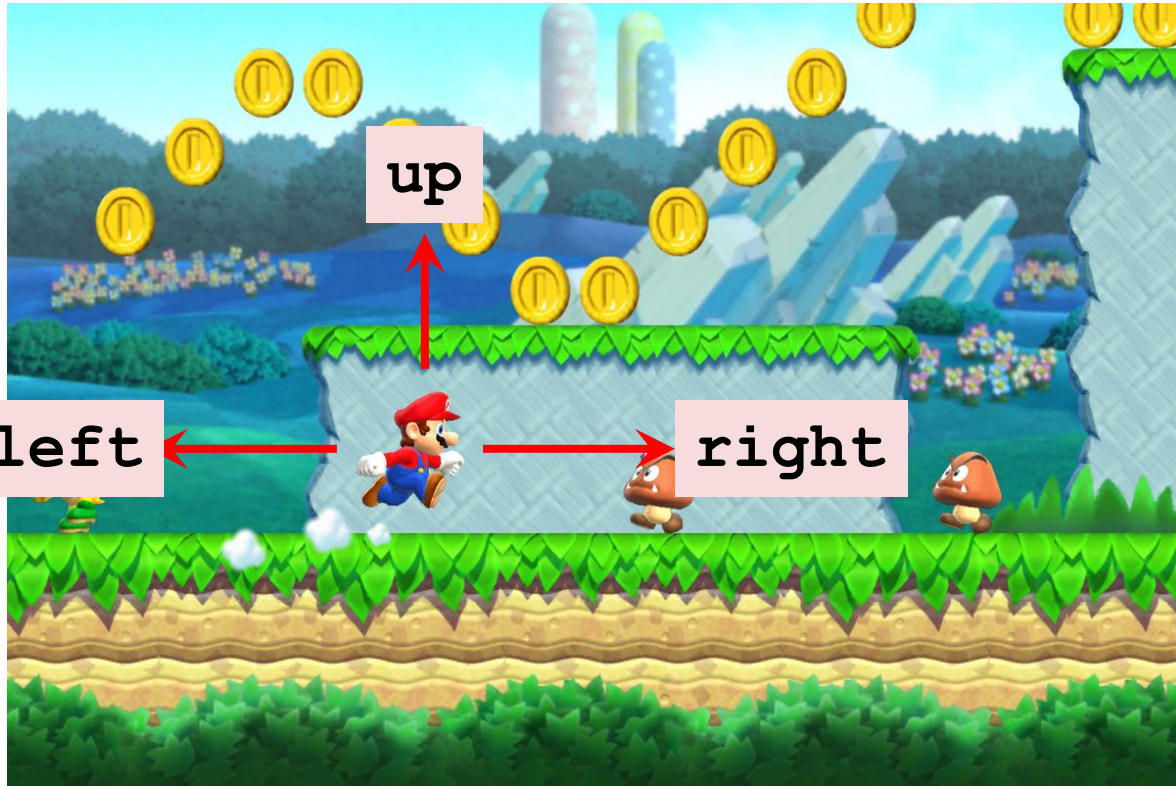
- Policy function $\pi: (s, a) \mapsto [0,1]$:

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Terminology: policy

policy π



- $\pi(a | s)$ is the probability of taking action $A = a$ given state s , e.g.,

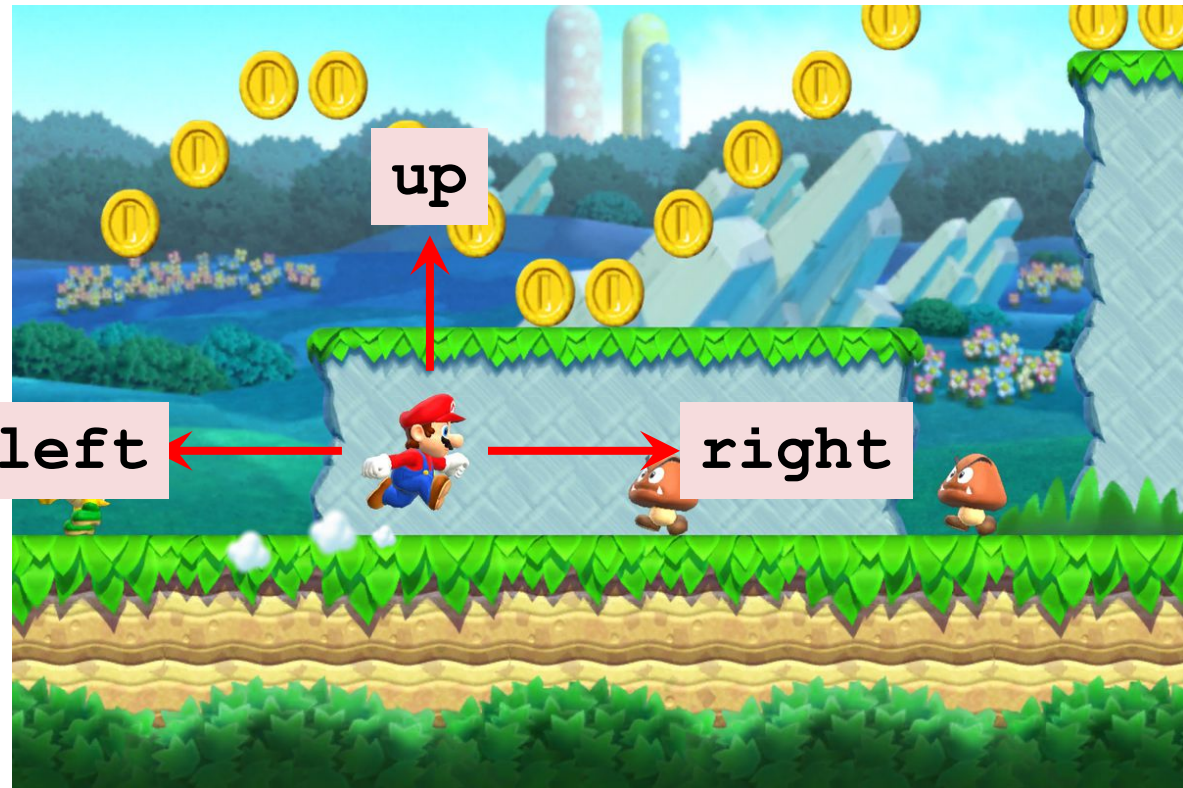
→ • $\pi(\text{left} | s) = 0.2,$

→ • $\pi(\text{right} | s) = 0.1,$

→ • $\pi(\text{up} | s) = 0.7.$

Terminology: policy

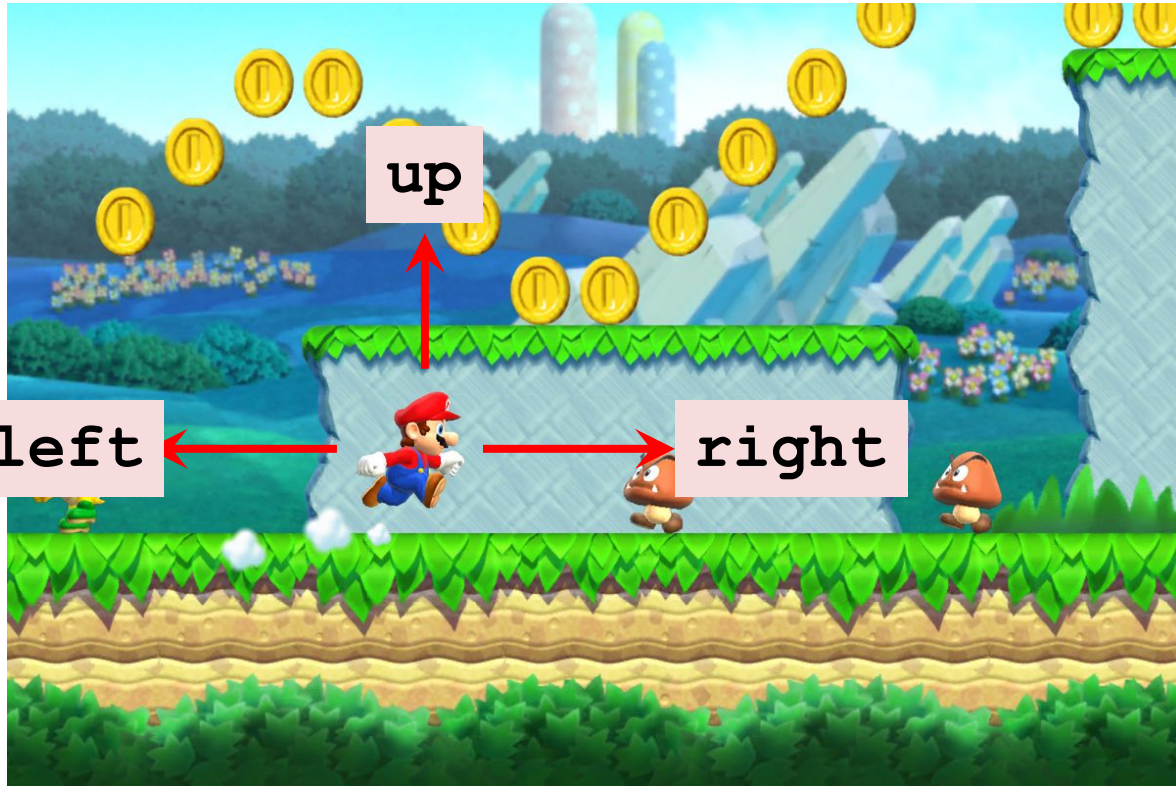
policy π



- $\pi(a | s)$ is the probability of taking action $A = a$ given state s , e.g.,
 - $\pi(\text{left} | s) = 0.2$,
 - $\pi(\text{right} | s) = 0.1$,
 - $\pi(\text{up} | s) = 0.7$.
- Upon observing state $S = s$, the agent's action A can be random.

Terminology: policy

Random or deterministic policy?



Terminology: reward

reward R

- Collect a coin: $R = +1$



Terminology: reward

reward R



- Collect a coin: $R = +1$
- Win the game: $R = +10000$

Terminology: reward

reward R



- Collect a coin: $R = +1$
- Win the game: $R = +10000$
- Touch a Goomba: $R = -10000$ (game over).

Terminology: reward

reward R



- Collect a coin: $R = +1$
- Win the game: $R = +10000$
- Touch a Goomba: $R = -10000$ (game over).
- Nothing happens: $R = 0$

Terminology: state transition

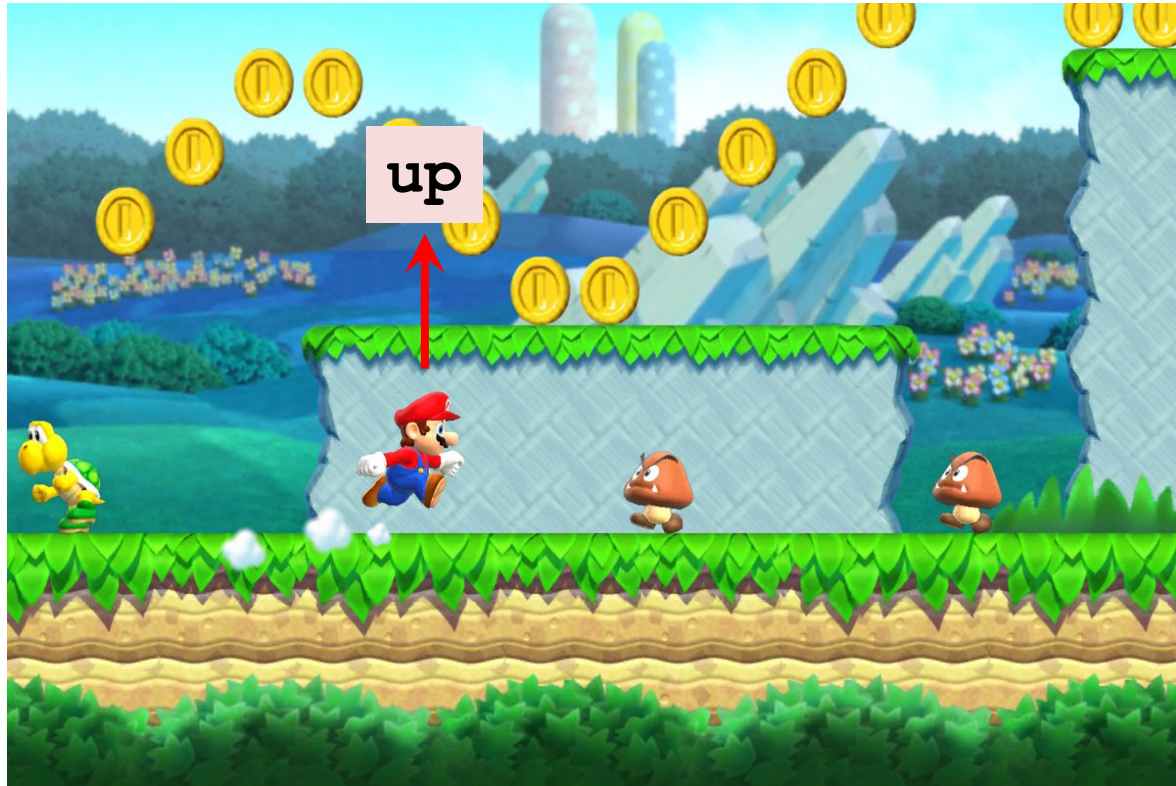


state transition



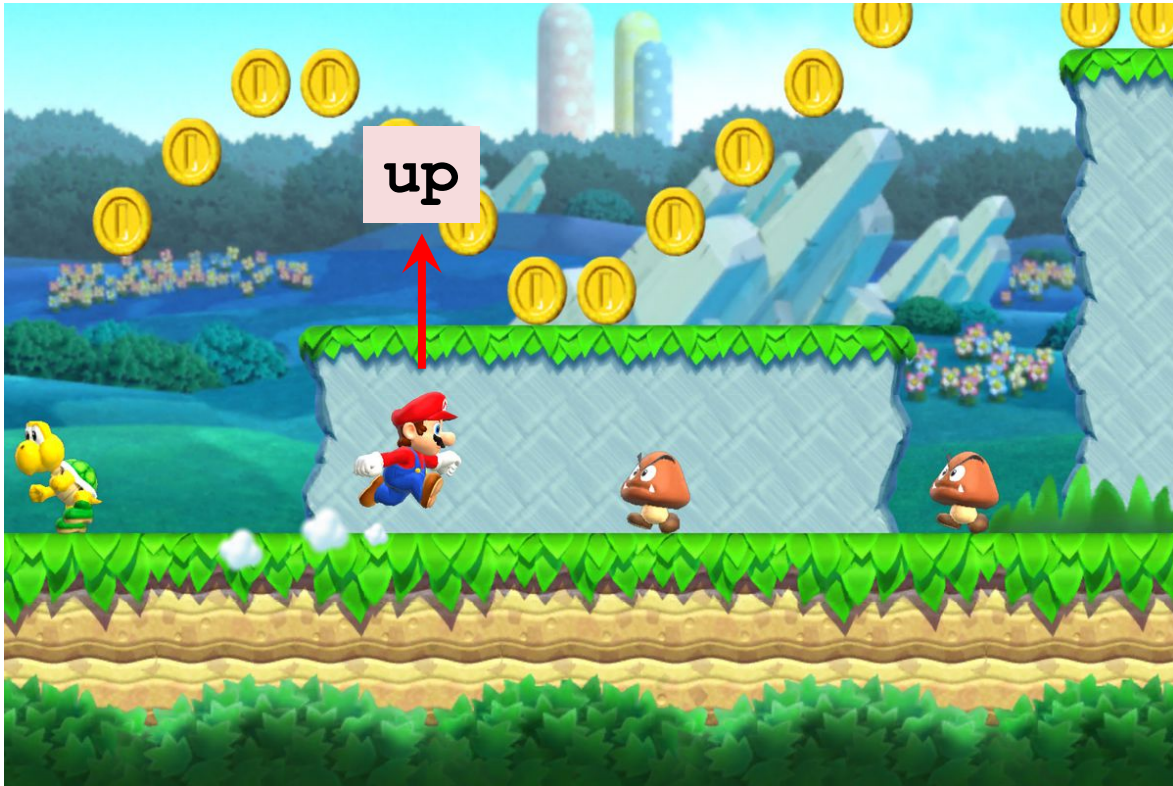
Terminology: state transition

state transition



- E.g., “up” action leads to a new state.

Terminology: state transition



state transition



- E.g., “up” action leads to a new state.
- State transition can be random.
- Randomness is from the environment.

Terminology: state transition

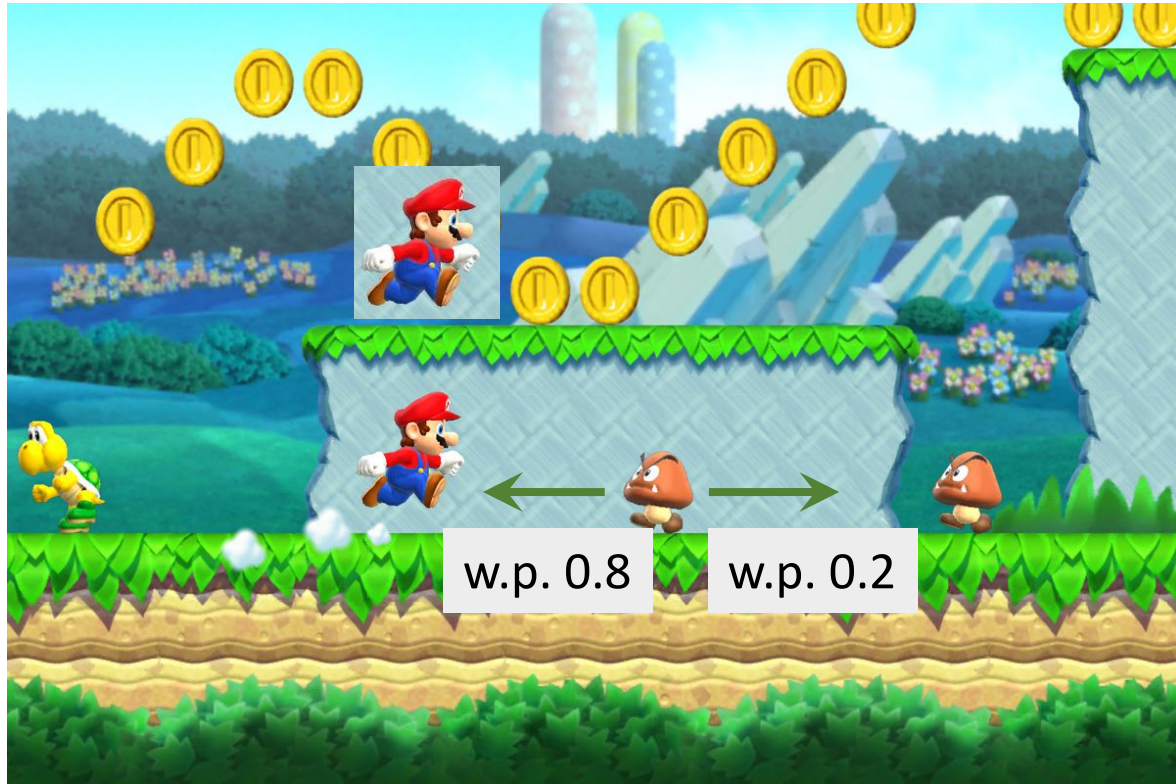


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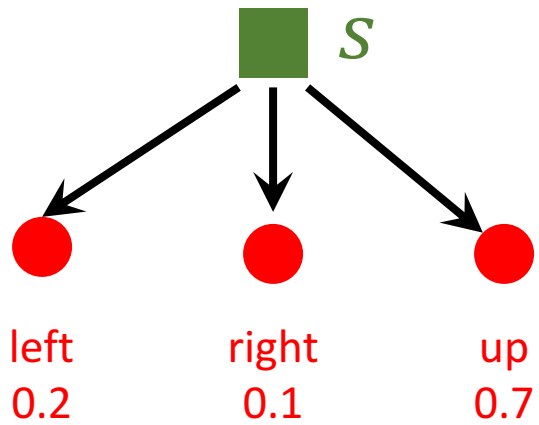
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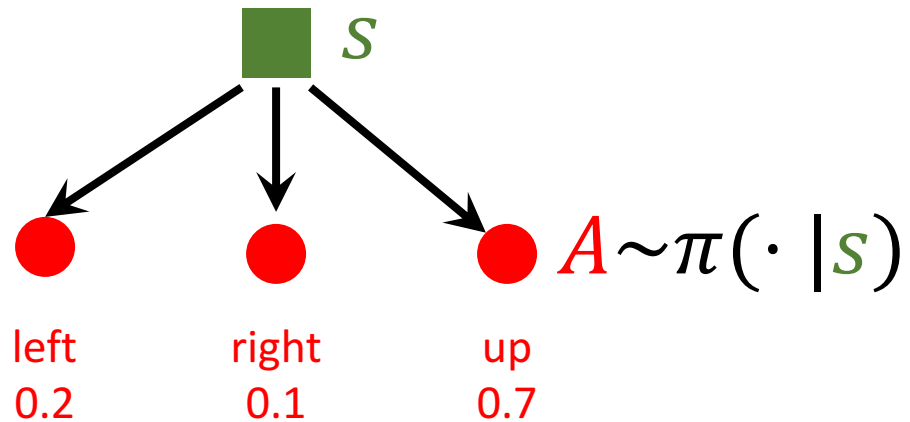
- E.g., “up” action leads to a new state.
- State transition can be random.
- Randomness is from the environment.
- $p(s'|s, a) = \mathbb{P}(S' = s' | S = s, A = a)$.

Two Sources of Randomness

Randomness in Actions



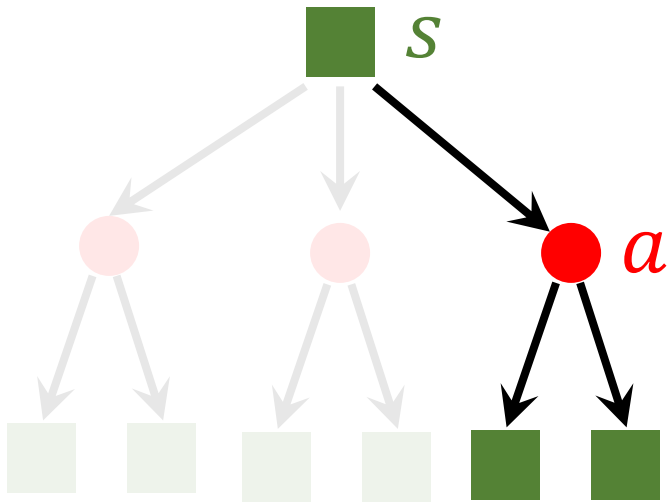
Randomness in Actions



Given state s , the action can be random, e.g., .

- $\pi(\text{"left"} | s) = 0.2,$
- $\pi(\text{"right"} | s) = 0.1,$
- $\pi(\text{"up"} | s) = 0.7.$

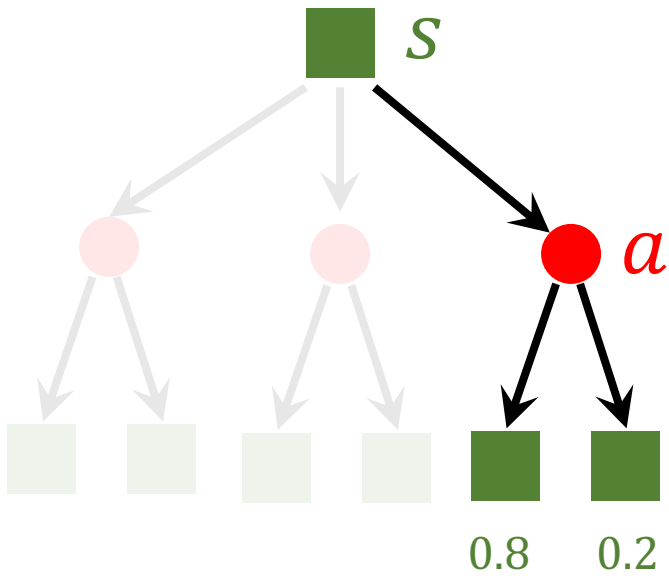
Randomness in States



- State transition can be random.
- The environment generates the new state s' by

$$\underline{s' \sim p(\cdot | s, a) .}$$

Randomness in States



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Two Sources of Randomness

- The randomness in **action** is from the policy function:

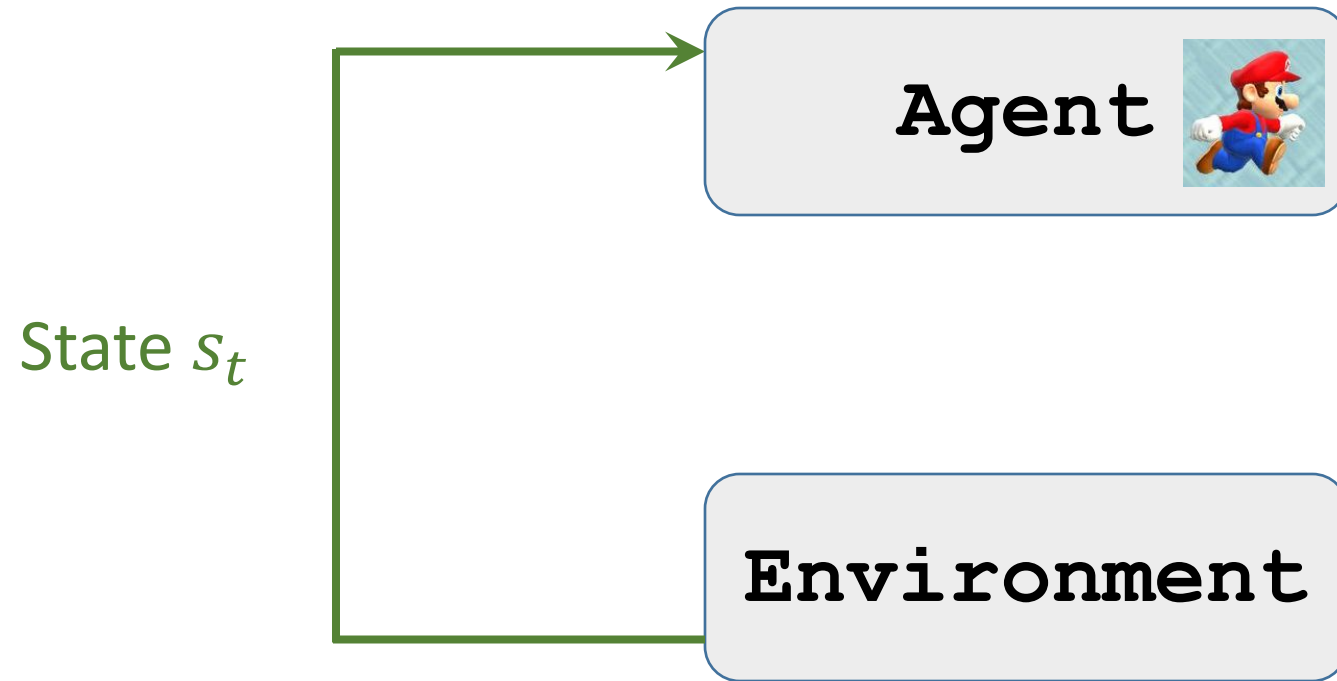
$$A \sim \pi(\cdot \mid s) .$$

- The randomness in **state** is from the state-transition function:

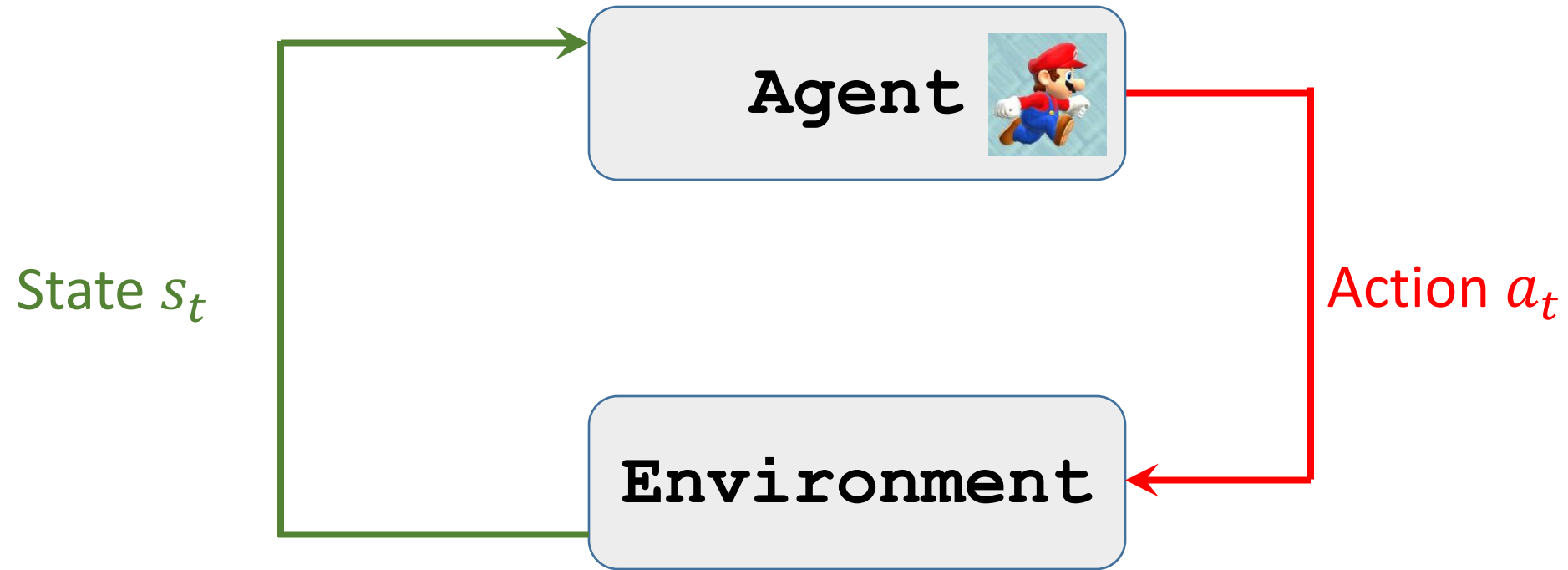
$$S' \sim p(\cdot \mid s, a) .$$

Agent-Environment Interaction

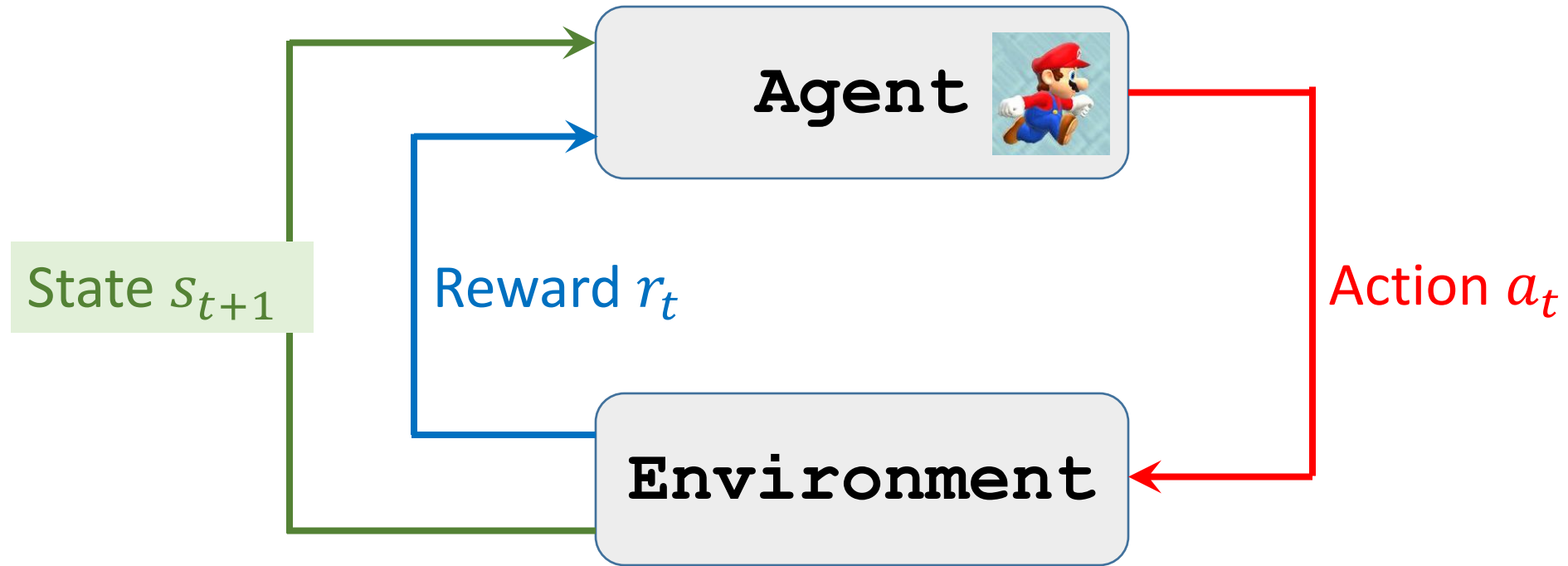
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Agent-Environment Interaction



Agent-Environment Interaction

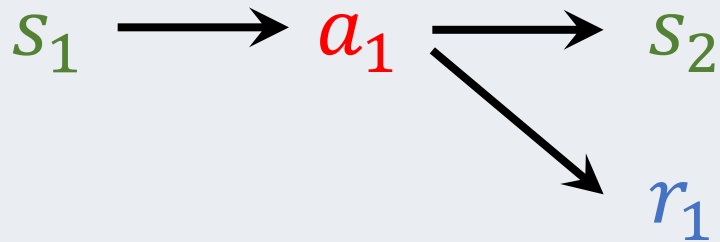


Play game using AI

- Observe state s_t , select action $a_t \sim \pi(\cdot \mid s_t)$, and execute a_t .
- The environment gives new state s_{t+1} and reward r_t .

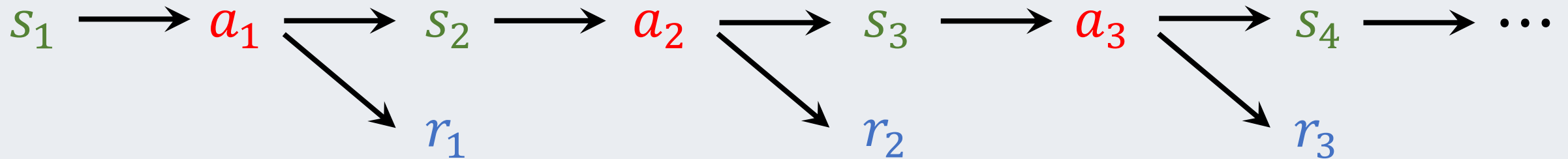
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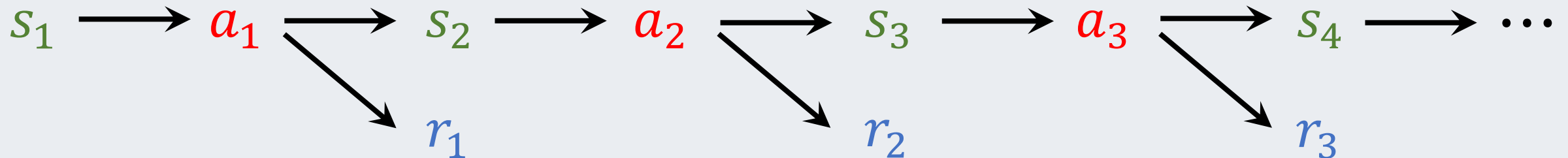


Play game using AI

- (state, action, reward) trajectory:

$s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_n, a_n, r_n$

- One episode is from the the beginning to the end (Mario wins or dies).



Play game using AI

- (state, action, reward) trajectory:

$$s_1, a_1, r_1, \quad s_2, a_2, r_2, \quad \cdots, \quad s_n, a_n, r_n.$$

- One episode is from the the beginning to the end (Mario wins or dies).
- What is a good policy?
- A good policy leads to big cumulative reward: $\sum_{t=1}^n \gamma^{t-1} \cdot r_t$.
- Use the rewards to guide the learning of policy.

Rewards and Returns

Return

Definition: Return (aka cumulative future reward).

- $\underline{U_t} = \underline{R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots}$

Return

Definition: Return (aka cumulative future reward).

- $U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots$

Question: At time t , are R_t and R_{t+1} equally important?

- Which of the followings do you prefer?
 - I give you \$100 right now.
 - I will give you \$100 one year later.

Return

Definition: Return (aka cumulative future reward).

- $U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots$

Question: At time t , are R_t and R_{t+1} equally important?

- Which of the followings do you prefer?
 - I give you \$80 right now.
 - I will give you \$100 one year later.

Return

Definition: Return (aka cumulative future reward).

- $U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots$

Question: At time t , are R_t and R_{t+1} equally important?

- Which of the followings do you prefer?
 - I give you \$80 right now.
 - I will give you \$100 one year later.
- Future reward is less valuable than present reward.
- R_{t+1} should be given less weight than R_t .

Discounted Returns

Definition: Return (aka cumulative future reward).

- $U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots$

Definition: Discounted return (aka cumulative discounted future reward).

- γ : discount factor (tuning hyper-parameter).

- $U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \gamma^3 R_{t+3} + \dots$

Discounted Returns

Definition: Discounted return (at time t).

- $U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$

Randomness in Returns

Definition: Discounted return (at time t).

- $U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$

At the end of the game, we observe u_t .

- We observe all the rewards, $r_t, r_{t+1}, r_{t+2}, \dots, r_n.$
- We thereby know the discounted return:

$$u_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^{n-t} r_n.$$

Randomness in Returns

Definition: Discounted return (at time t).

- $U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$

At time t , the rewards, $R_t, \dots, R_n$, are **random**, so the return U_t is **random**.

Randomness in Returns

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At time t , the rewards, $R_t, \dots, R_n$, are **random**, so the return U_t is **random**.

- Reward R_i depends on S_i and A_i .
- States can be random: $S_i \sim p(\cdot \mid s_{i-1}, a_{i-1}).$
- Actions can be random: $A_i \sim \pi(\cdot \mid s_i).$
- If either S_i or A_i is random, then R_i is random.

Randomness in Returns

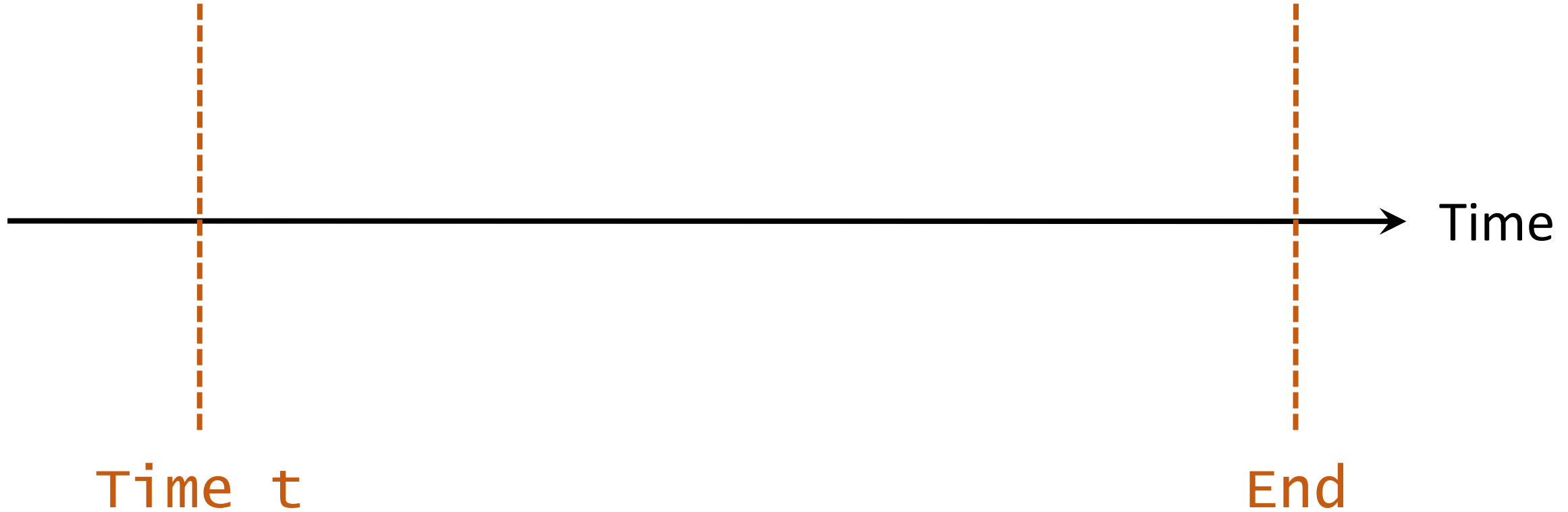
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At time t , the rewards, $R_t, \dots, R_n$, are **random**, so the return U_t is **random**.

- Reward R_i depends on S_i and A_i .
- U_t depends on $R_t, R_{t+1}, \dots, R_n$.
- $\Rightarrow U_t$ depends on $S_t, A_t, S_{t+1}, A_{t+1}, \dots, S_n, A_n$.

Randomness in Returns



Randomness in Returns



R_t

R_{t+1}

R_{t+2}

...

R_n

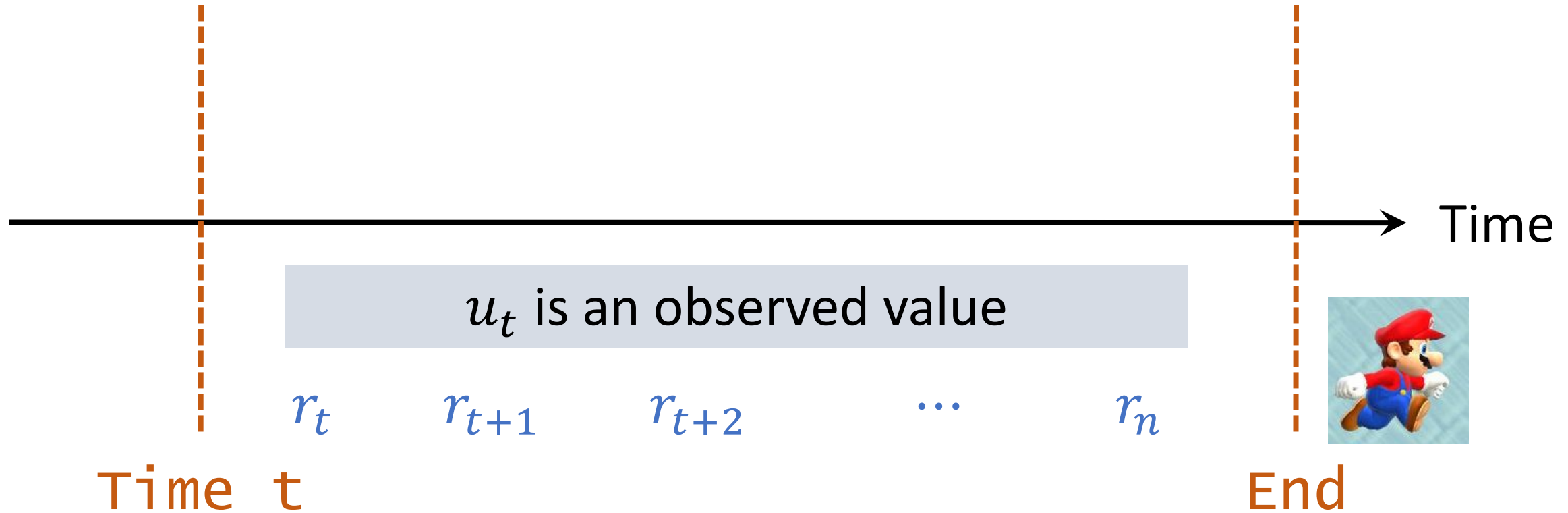
U_t is a random variable

Time

Time t

End

Randomness in Returns



Value Functions

Action-Value Function $Q_{\pi}(s, a)$

Definition: Discounted return.

- $U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$

Action-Value Function $Q_{\pi}(s, a)$

Definition: Discounted return.

- $U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$

Definition: Action-value function.

- $Q_{\pi}(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$

Action-Value Function $Q_{\pi}(s, a)$

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Definition: Action-value function.

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U_t depends on states $s_t, s_{t+1}, \dots, s_n$ and actions $a_t, a_{t+1}, \dots, a_n$.

Action-Value Function $Q_\pi(s, a)$

Definition: Discounted return.

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Regard s_t and a_t as observed values.

Regard $\underline{s_{t+1}, \dots, s_n}$ and $\underline{A_{t+1}, \dots, A_n}$ as random variables.

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Action-Value Function $Q_{\pi}(s, a)$

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Definition: Action-value function.

$$\bullet Q_{\pi}(s_t, a_t) = \mathbb{E}[U_t \mid S_t = s_t, A_t = a_t].$$

$$\begin{aligned} \bullet S_{t+1} &\sim p(\cdot \mid s_t, a_t), \\ &\vdots \\ \bullet S_n &\sim p(\cdot \mid s_{n-1}, a_{n-1}). \end{aligned}$$

$$\begin{aligned} \bullet A_{t+1} &\sim \pi(\cdot \mid s_{t+1}), \\ &\vdots \\ \bullet A_n &\sim \pi(\cdot \mid s_n). \end{aligned}$$

Action-Value Function $Q_{\pi}(s, a)$

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Action-Value Function $Q_{\pi}(s, a)$

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Definition: Action-value function.

- $Q_{\pi}(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$

- $Q_{\pi}(s_t, a_t)$ depends on s_t, a_t, π , and p .

- $Q_{\pi}(s_t, a_t)$ is dependent of $\underline{s_{t+1}, \dots, s_n}$ and $\underline{A_{t+1}, \dots, A_n}$.

State-Value Function $V_{\pi}(s)$

Definition: Discounted return.

- $U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$

Definition: Action-value function.

- $Q_{\pi}(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$

Definition: State-value function.

- $\underline{V_{\pi}(s_t)} = \mathbb{E}_{\underline{A}} [\underline{Q_{\pi}(s_t, A)}]$

State-Value Function $V_{\pi}(s)$

Definition: Discounted return.

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Definition: State-value function.

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Taken w.r.t. the action $A \sim \pi(\cdot \mid s_t).$

State-Value Function $V_{\pi}(s)$

Definition: Discounted return.

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Understanding the Value Functions

- **Action**-value function: $\underline{Q_\pi(s, a)} = \mathbb{E} [\underline{U_t | S_t = s, A_t = a}]$.
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- $\mathbb{E}_S [V_{\pi}(S)]$ evaluates how good the policy π is.

Evaluating Reinforcement Learning

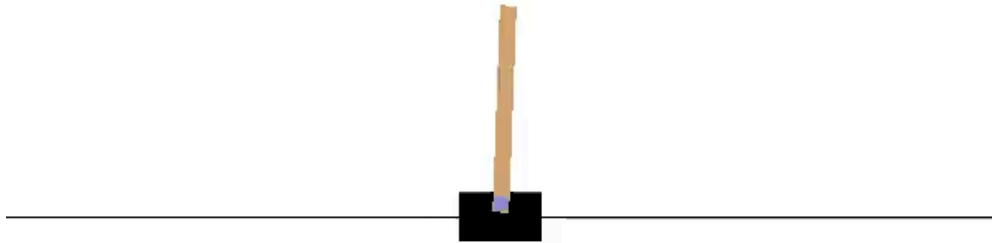
OpenAI Gym

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Classical control problems



Cart Pole

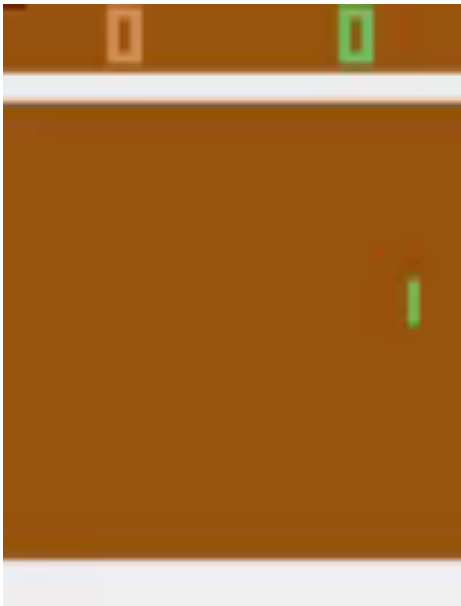


Pendulum

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Atari Games



Pong



Space Invader

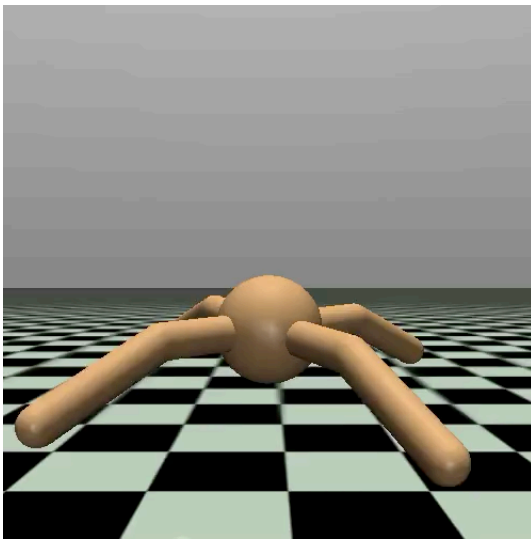


Breakout

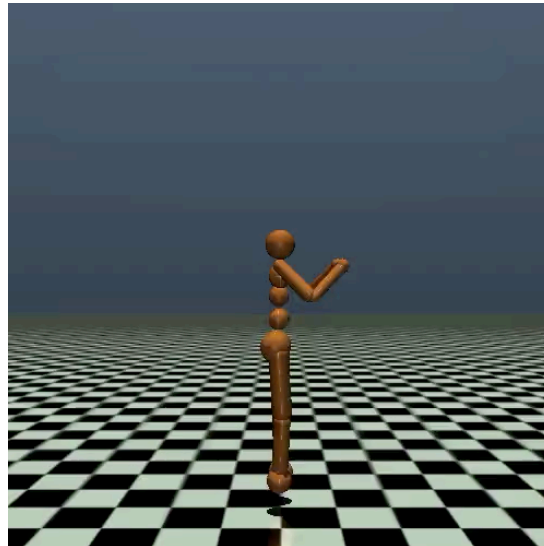
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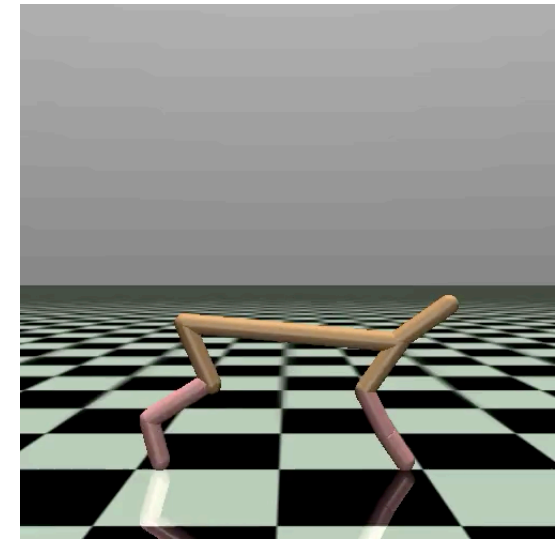
MuJoCo (Continuous control tasks.)



Ant

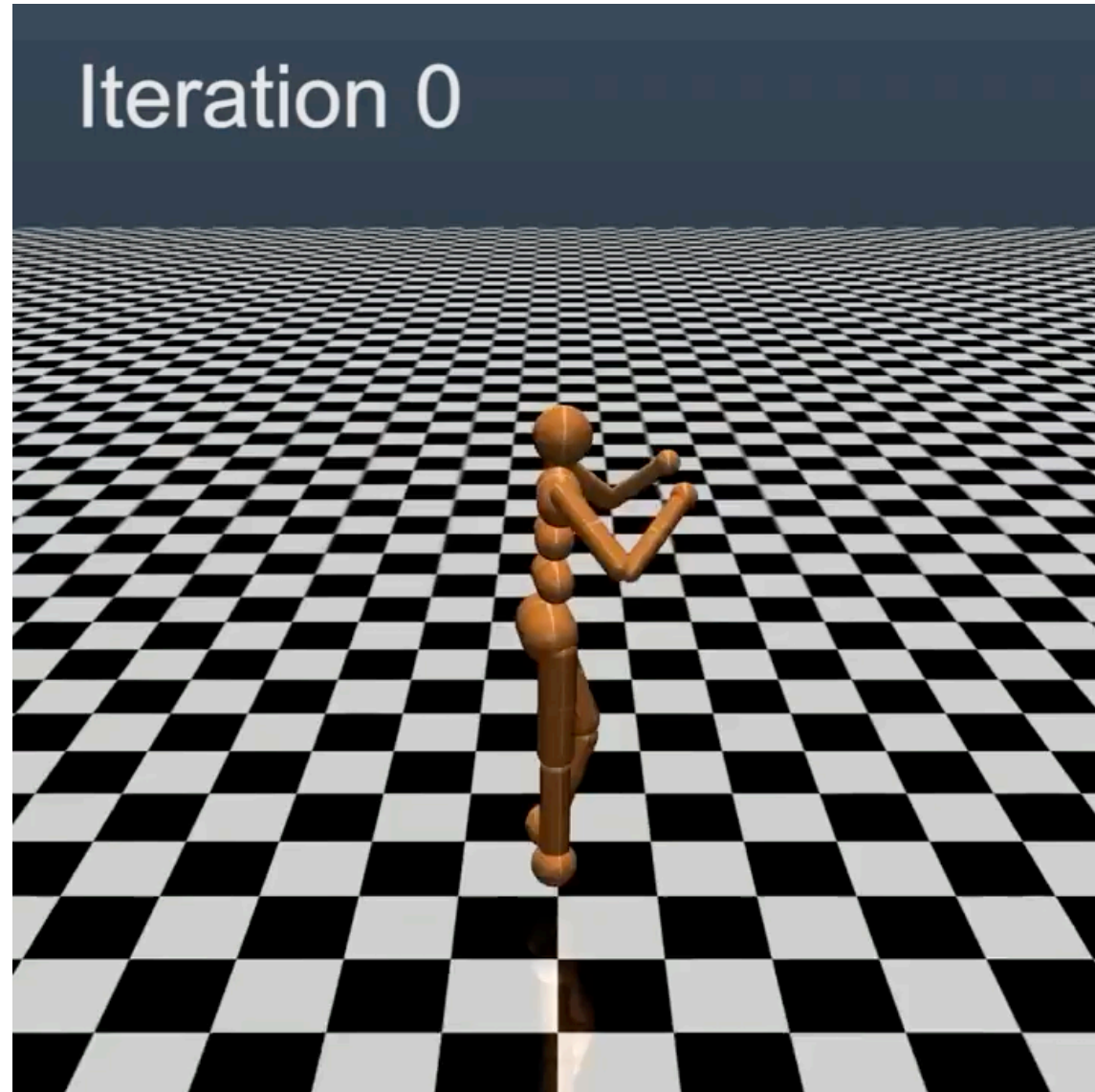


Humanoid



Half Cheetah

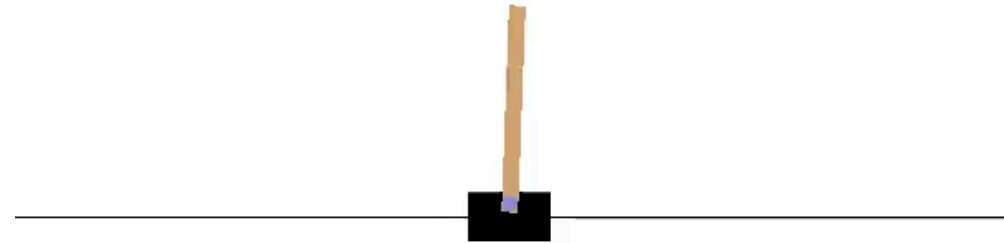
OpenAI Gym



Play CartPole Game

```
➡ import gym  
➡ env = gym.make( 'CartPole-v0' )
```

- Get the environment of CartPole from Gym.
- “env” provides states and reward.



Play CartPole Game

➡ `state = env.reset()`

➡ `for t in range(100):` ➡ A window pops up rendering CartPole.

➡ `env.render()`
`print(state)`

A random **action**.

➡ `action = env.action_space.sample()`

➡ `state, reward, done, info = env.step(action)`

➡ `if done:` "done=1" means finished (win or lose the game)

`print('Finished')`

`break`

`env.close()`

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- Environment
- State s
- Action a
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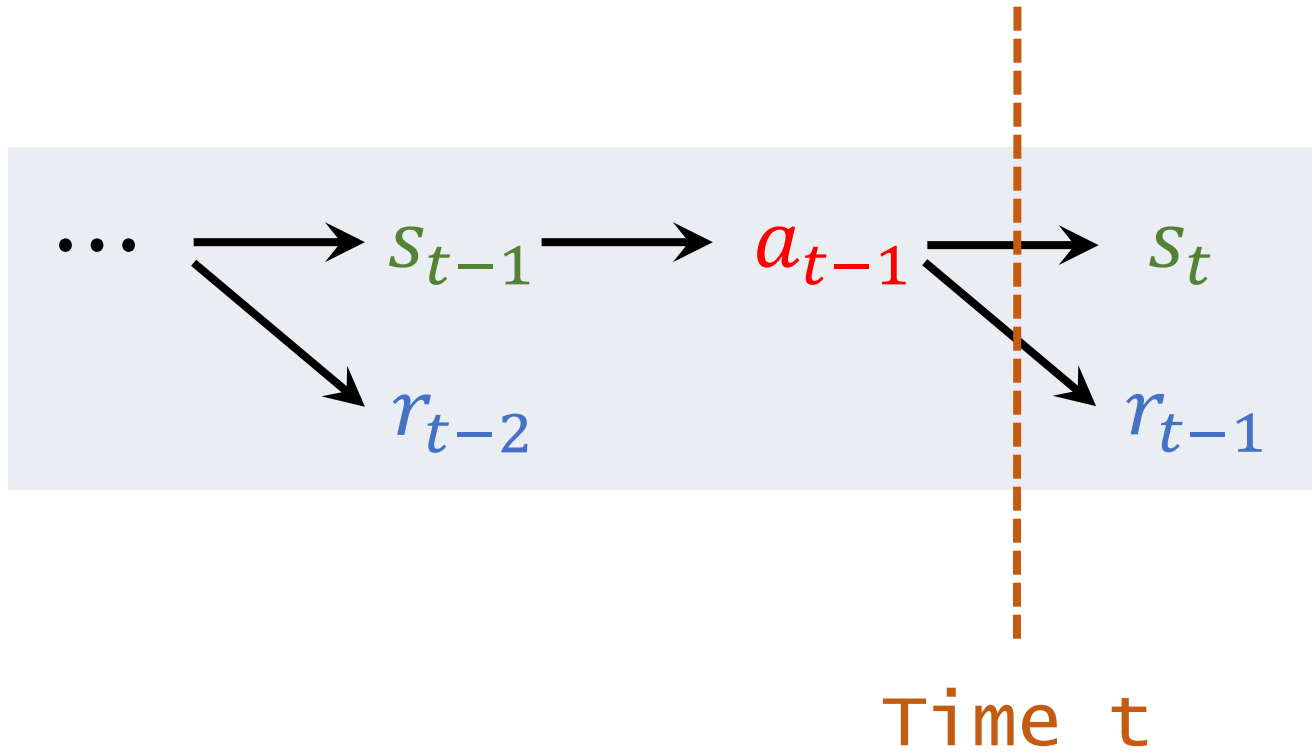
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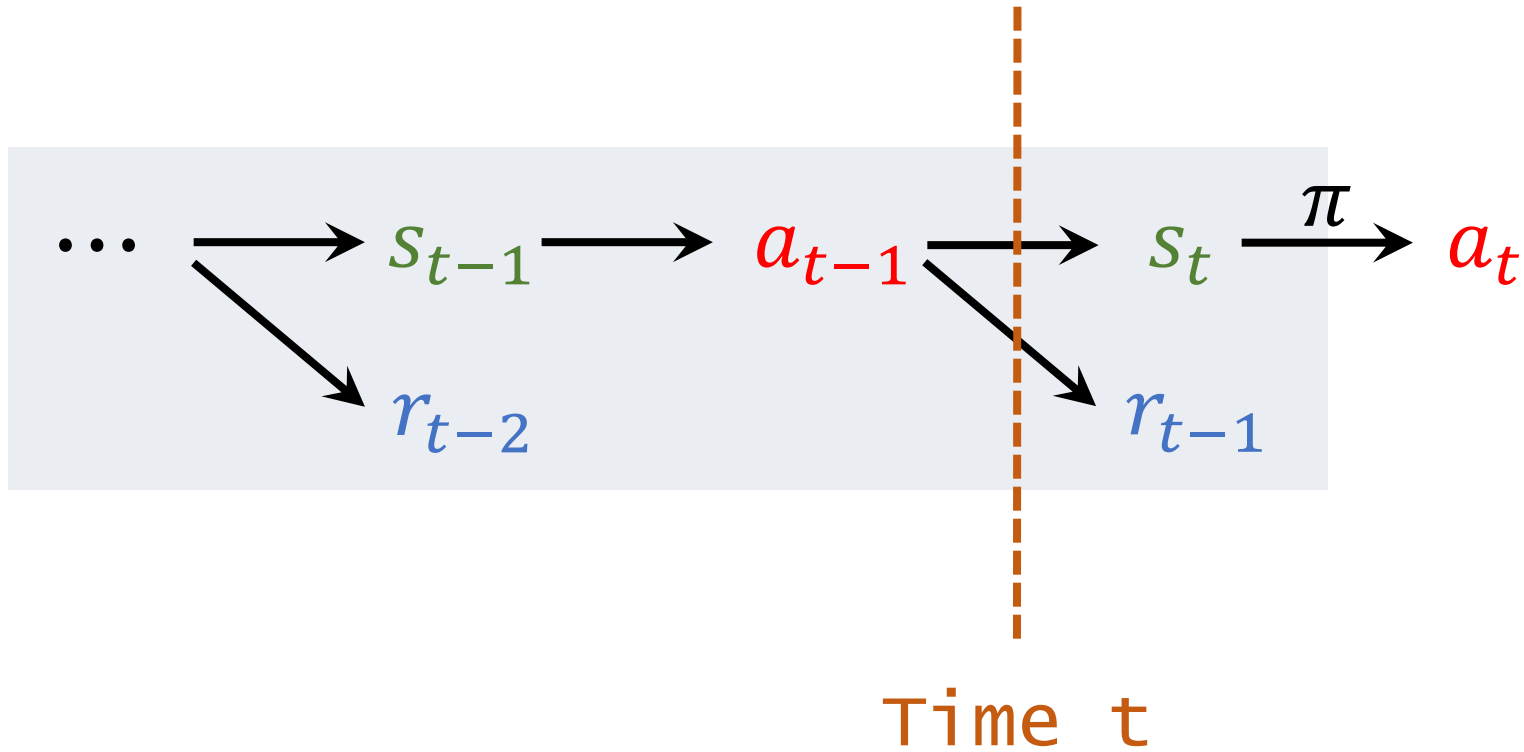
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$$V_\pi(s_t) = \mathbb{E}_{A \sim \pi}[Q_\pi(s_t, A)].$$

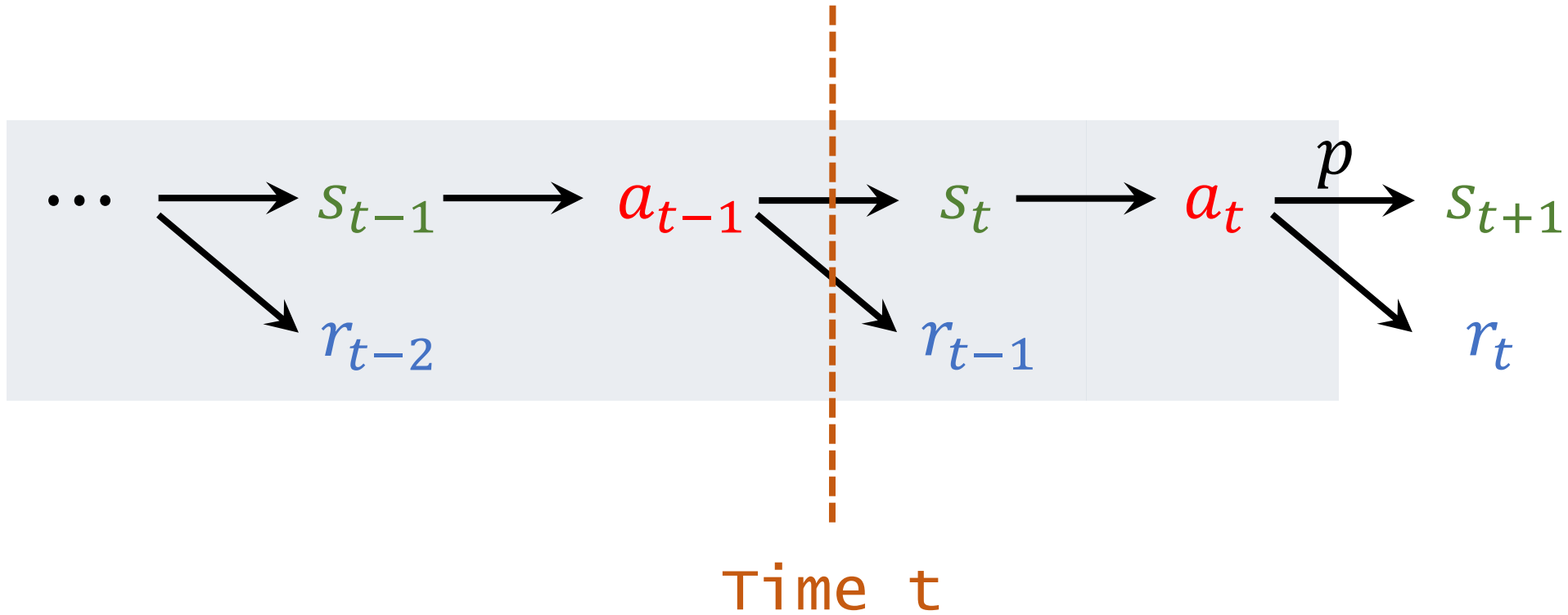
Play game using AI



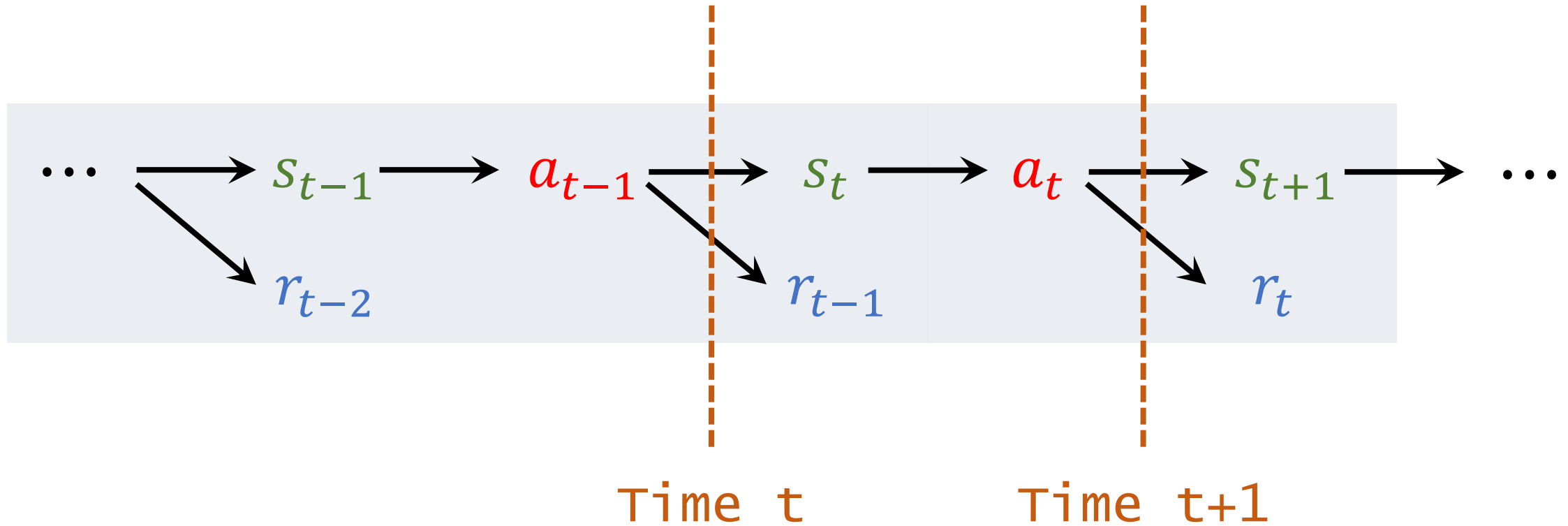
Play game using AI



Play game using AI



Play game using AI



Thank You!

<http://wangshusen.github.io/>